

Resilient Distributed Quadrotor AAVs Game: Addressing FDI and Physical Attacks

Mei-Jie Geng , Hua-Feng Ding , Xiang-Yu Yao , and Maolong Lv 

Abstract—Distributed quadrotor autonomous aerial vehicles (AAVs) are increasingly deployed in critical applications, including reconnaissance, disaster response, and transportation, where system resilience is paramount. However, their complex dynamics and reliance on cyber-physical systems make them highly vulnerable to adversities such as uncertain dynamics, time-varying input disturbances, false data injection (FDI), and physical attacks. This paper proposes a hierarchical framework consisting of a game layer and an execution layer to enhance the resilience of a noncooperative game involving distributed AAVs. In the game layer, a novel FDI observer mitigates communication and computational FDI attacks, facilitating convergence to infinitesimal ranges around the Nash equilibrium (NE) point. The execution layer addresses uncertain dynamics and physical attacks, ensuring that AAVs can track a narrow neighborhood of the NE even in the presence of actuator failures. Extensive simulations validate the algorithm's effectiveness and robustness against mixed attack scenarios, underscoring its potential for real-world AAV applications.

Index Terms—Noncooperative game, quadrotor (AAVs), false data injection attacks, physical attacks.

I. INTRODUCTION

IN RECENT years, distributed quadrotor AAVs have gained significant popularity across various fields such as reconnaissance, search and rescue, and transportation, owing to their versatility, compactness, and affordability. Extensive research has been conducted on different aspects of distributed quadrotor AAVs, including formation control [1], [2], [3], [4], tracking [6], [7], [8], flocking [9], [10], [11], and other related control issues [5], [12]. However, as application scenarios become more complex and challenging, there is a growing demand

to enhance the thinking and decision-making capabilities of quadrotor AAVs.

Game theory and optimization have recently garnered significant attention in the field of automation. In [13], a method for seeking anti-disturbance NE in games with partial information was proposed. For linear systems, a pseudo-gradient method was introduced to seek NE in [14]. Authors in [15] studied time-varying optimization and proposed finite-time convergent algorithms for simple first-order systems. Zhao et al. [16] investigated prescribed-time NE seeking for first-order noncooperative game players. It's evident that game theory can provide a powerful framework for addressing coordination and decision-making challenges in multi-agent systems. NE is commonly adopted due to its clear definition and mathematical tractability, and other game-theoretic concepts, such as mixed strategy equilibria and evolutionary stable strategies (ESS), offer alternative approaches to model agent interactions in more complex or dynamic environments. Mixed strategy equilibria can allow agents to adopt probabilistic strategies to optimize their outcomes, which is particularly useful in scenarios where deterministic strategies are insufficient. On the other hand, ESS, originating from evolutionary game theory, provides a robust criterion for stability in adaptive systems and can capture the strategic evolution of AAV behaviors under varying environmental conditions. These concepts have been applied to AAV applications, such as task assignment and conflict resolution [31], [32], while they often lack rigorous mathematical proof and comprehensive stability analysis in AAV systems. This limitation hinders their scalability and practical deployment in large-scale systems. Although numerous game-theoretic methods have been developed to address various gaming scenarios, the search for NE remains one of the most classical and effective strategies. It offers a robust mathematical foundation upon which more advanced game-theoretic approaches can be built. Furthermore, most of the works have only focused on research involving simple systems. However, quadrotor AAVs are complex systems with inner and outer loops, and there is currently limited research on game theory and intelligent decision-making processes specifically tailored for distributed quadrotor AAVs. Further research is needed to explore how to integrate game theory concepts into the flight of quadrotor AAVs to enhance their intelligence. This involves investigating strategies for decision-making, coordination, and adaptation based on game-theoretic principles to improve system overall performance and capabilities.

As a typical cyber-physical system (CPS), quadrotor AAVs must address security concerns in both their cyber and physical

Received 5 December 2024; revised 11 February 2025; accepted 25 March 2025. Date of publication 31 March 2025; date of current version 15 August 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62303432 and Grant 62233007, in part by the Key Laboratory of System Control and Information Processing Foundation under Grant Scip20230102, and in part by the Hubei Provincial Natural Science Foundation of China under Grant 2023AFB038. The review of this article was coordinated by Prof. Tiago Koketsu Rodrigues. (Corresponding authors: Hua-Feng Ding; Xiang-Yu Yao.)

Mei-Jie Geng and Hua-Feng Ding are with the School of Mechanical Engineering and Electronic Information, China University of Geosciences, Wuhan 430074, China (e-mail: Gmajor@cug.edu.cn; dinghf@cug.edu.cn).

Xiang-Yu Yao is with the School of Mechanical Engineering and Electronic Information, China University of Geosciences, Wuhan 430074, China, and also with the Key Laboratory of System Control and Information Processing, Ministry of Education, Shanghai 200240, China (e-mail: xyyao518@163.com).

Maolong Lv is with the College of Air Traffic Control and Navigation, Air Force Engineering University, Xi'an 710038, China (e-mail: maolonglv@163.com).

Digital Object Identifier 10.1109/TVT.2025.3556228

layers during flight. In complex environments, quadrotor AAVs are susceptible to external false data injection (FDI) attacks affecting both their communication and computational aspects in the cyber layer. Without adequate measures to mitigate these attacks, their ability to perform flight missions is successfully compromised. In [17], the authors considered FDI attacks in nonlinear CPS and proposed an event-triggered fixed-time control approach directly applicable to quadrotor AAV flight control. In [18], a modified sliding innovation sequences method was used to detect and prevent FDI attacks. A reinforcement learning approach for handling FDI attacks during cooperative operations of quadrotor AAVs was mentioned in the review article [19]. In [20], dynamic reconstitution and secure control problems regarding FDI attacks were investigated, with an adjustable proportional-integral multivariable observer designed to resist actuator and sensor FDI attacks. Regarding physical layer security for actuator operation, quadrotor AAVs should employ measures to detect and mitigate potential physical attacks that could compromise actuator functionality or safety [21], [22], [23], [24]. Physical attacks primarily result in damage to propellers or degradation of motor performance in quadrotor AAVs. Wang et al. [25] proposed a real-time fault detection method with a model-based acceleration engine. Note that fault-tolerant methods can be utilized to address actuator failures caused by physical attacks [26]. In [27], a neural network method was employed to address actuator faults for a quadrotor AAV. Self-scheduled fault-tolerant control laws were designed for a multicopter AAV in [28]. Existing works focus on addressing specific types of attacks, such as physical attacks or communication disturbances, but rarely consider mixed attack scenarios. These approaches often fail to provide a unified framework for handling diverse threats simultaneously. Moreover, traditional fault-tolerant strategies often depend on centralized architectures, which are impractical for scalable distributed AAV systems in dynamic environments. Given the importance of flight security and safety in cyber and physical layers of quadrotor AAVs, it is crucial to simultaneously address challenges posed by complex and aggressive composite attacks. Therefore, it is essential to simultaneously consider the impact of FDI and physical attacks to ensure the safety and security of distributed quadrotor AAVs system operations.

Building upon the preceding discussions, this paper investigates the resilient noncooperative game of distributed quadrotor AAVs amidst FDI and physical attacks. The novel contributions are summarized as follows:

- 1) Existing literature lacks effective methods to address FDI attacks faced by quadrotor AAVs in noncooperative games. To bridge this gap, we propose a novel FDI observer capable of countering compound FDI attacks in both communication and computation for quadrotor AAVs engaged in noncooperative games. This observer enables all quadrotor AAVs to converge arbitrarily close to the NE point and ensures the overall stability and robustness of the distributed system operating within the noncooperative game.
- 2) For comprehensive system stability analysis, we employ a hierarchical framework that facilitates individual

management of complex hybrid attacks. The distributed system is divided into a game layer and an execution layer. In the game layer, FDI attacks are effectively mitigated using a newly developed observer. Meanwhile, the execution layer is dedicated to addressing physical attacks while also managing system uncertainties and time-varying input disturbances concurrently, ensuring system robustness. This hierarchical framework enables thorough analysis and mitigation of various attack types in distributed quadrotor AAVs systems.

II. PRELIMINARIES AND FORMULATION

A. Graph Theory

In the system, the communication network of the quadrotor AAVs is described using an undirected graph. The graph is represented as $\mathbb{G} = \{\mathbb{V}, \mathbb{E}, \mathbb{A}\}$, where $\mathbb{V} = \{1, \dots, n\}$ denotes the vertex set, and $\mathbb{E} \subseteq \mathbb{V} \times \mathbb{V}$ represents the edge set. An edge $(\mathbb{V}_i, \mathbb{V}_j) \in \mathbb{E}$ indicates that quadrotor AAV- i and quadrotor AAV- j can communicate directly. The set of adjacent nodes of quadrotor AAV- i is denoted as $\mathcal{N}_i = \{j \in \mathbb{V} | (i, j) \in \mathbb{E}\}$. The adjacency matrix of the graph $\mathbb{G}$ is denoted as $\mathcal{A} = [a_{ij}] \in \mathbb{R}^{n \times n}$. In $\mathcal{A}$, if $i = j$, $a_{ij} = 0$, and if $(\mathbb{V}_i, \mathbb{V}_j) \in \mathbb{E}$, $a_{ij} > 0$, otherwise, $a_{ij} = 0$. The Laplacian matrix of the undirected graph $\mathbb{G}$ is defined as $\mathcal{L} = \mathcal{D} - \mathcal{A}$, where $\mathcal{D} = \text{diag}\{d_1 \dots d_n\}$ is a diagonal matrix with $d_i = \sum_{j=1}^n a_{ij}$. It is worth noting that $\mathcal{L}1_n = 0_n$. The eigenvalues of the matrix $\mathcal{L}$ are denoted by $\lambda_i > 0$, where $i \in \{1, \dots, n\}$.

Assumption 1: Each quadrotor AAV can interact with other agents through undirected and connected graph $\mathbb{G}$.

B. Game Theory

In the noncooperative game of n quadrotor AAVs, the cost function of each agent is $\mathbb{J}_i(q_i, q_{-i})$, where q_i is its own output state and $q_{-i} = \text{col}\{q_1, \dots, q_{i-1}, q_{i+1}, \dots, q_n\}$ represents the output state of other agents. Assuming the existence of an NE in the game, develop a strategy aimed at seeking the NE that quadrotor AAVs can adopt to learn the optimal outcome of the noncooperative game.

Definition 1: The whole quadrotor AAVs system reaches the NE when all quadrotor AAVs get the minimized cost function $\mathbb{J}_i(q_i^*, q_{-i}^*)$.

$$\mathbb{J}_i(q_i^*, q_{-i}^*) \leq \mathbb{J}_i(q_i, q_{-i}^*), \forall i \in \mathcal{N}. \quad (1)$$

Assumption 2: The continuously differentiable cost function $\mathbb{J}_i(q_i, q_{-i})$ is convex in q_i for every fixed q_{-i} .

Assumption 3: $G(\mathbf{q}) = [\nabla_{q_1} \mathbb{J}_1(q_1, q_{-1}), \dots, \nabla_{q_n} \mathbb{J}_n(q_n, q_{-n})]^T$ denotes pseudo-gradient and it is ω -strongly monotone and ϕ -Lipschitz in $\mathbf{q}$.

Lemma 1: Based on Assumptions 2–3, $\mathbf{q}^* = [q_1^*, \dots, q_n^*]^T$ is a unique NE of the noncooperative game if

$$\nabla_{q_i} \mathbb{J}_i(q_i^*, q_{-i}^*) = 0, \forall i \in \mathbb{V}, \quad (2)$$

where $\nabla_{q_i} \mathbb{J}_i(q_i, q_{-i}) \in \mathbb{R}$ is the gradient of $\mathbb{J}_i$ at q_i for every fixed q_{-i} .

Additionally, for $x_1, x_2 \in \mathbb{R}^n$ with Assumption 3, we have

$$(x_1 - x_2)^T (G(x_1) - G(x_2)) > \omega \|x_1 - x_2\|^2. \quad (3)$$

Assumption 4: [29] When $\frac{\partial^2 \mathbb{J}_i}{\partial q_i^2}(\mathbf{q}^*) = 0$ and $\frac{\partial^2 \mathbb{J}_i}{\partial q_i^2}(\mathbf{q}^*) < 0$ are satisfied, at least one NE point exists, with the possibility of multiple equilibrium points.

Assumption 5: [29], [38], [39] The matrix

$$H = \begin{bmatrix} \frac{\partial^2 \mathbb{J}_1}{\partial q_1^2}(\mathbf{q}^*) & \frac{\partial^2 \mathbb{J}_1}{\partial q_1 \partial q_2}(\mathbf{q}^*) & \cdots & \frac{\partial^2 \mathbb{J}_1}{\partial q_1 \partial q_n}(\mathbf{q}^*) \\ \frac{\partial^2 \mathbb{J}_2}{\partial q_2 \partial q_1}(\mathbf{q}^*) & \frac{\partial^2 \mathbb{J}_2}{\partial q_2^2}(\mathbf{q}^*) & & \frac{\partial^2 \mathbb{J}_2}{\partial q_2 \partial q_n}(\mathbf{q}^*) \\ \vdots & & \ddots & \\ \frac{\partial^2 \mathbb{J}_n}{\partial q_n \partial q_1}(\mathbf{q}^*) & \frac{\partial^2 \mathbb{J}_n}{\partial q_n \partial q_2}(\mathbf{q}^*) & & \frac{\partial^2 \mathbb{J}_n}{\partial q_n^2}(\mathbf{q}^*) \end{bmatrix}$$

is strictly diagonally dominant, This means that $|\frac{\partial^2 \mathbb{J}_i}{\partial q_i^2}(\mathbf{q}^*)| > \sum_{j=1, j \neq i}^n |\frac{\partial^2 \mathbb{J}_i}{\partial q_i \partial q_j}(\mathbf{q}^*)|, \forall i \in n$.

C. Quadrotor AAVs Modeling

For each quadrotor AAV in the system, its dynamics model can be expressed as

$$\ddot{q}_{i,x} = (\cos \psi_i \sin \theta_i \cos \phi_i + \sin \psi_i \sin \phi_i) c_1 \omega_{i,s1} / m_i, \quad (4a)$$

$$\ddot{q}_{i,y} = (\sin \psi_i \sin \theta_i \cos \phi_i + \cos \psi_i \sin \phi_i) c_1 \omega_{i,s1} / m_i, \quad (4b)$$

$$\ddot{q}_{i,z} = (\cos \theta_i \cos \phi_i) c_1 \omega_{i,s1} / m_i - g, \quad (4c)$$

$$\ddot{\phi}_i = ((I_{yy} - I_{zz}) \dot{\theta}_i \dot{\psi}_i + c_1 l \omega_{i,s2}) / I_{xx}, \quad (4d)$$

$$\ddot{\theta}_i = ((I_{zz} - I_{xx}) \dot{\phi}_i \dot{\psi}_i + c_1 l \omega_{i,s3}) / I_{yy}, \quad (4e)$$

$$\ddot{\psi}_i = ((I_{xx} - I_{yy}) \dot{\phi}_i \dot{\theta}_i + c_2 \omega_{i,s4}) / I_{zz}. \quad (4f)$$

where $q_{i,x}, q_{i,y}$ and $q_{i,z}$ denote generalized spatial coordinates, ϕ_i, θ_i and ψ_i denote Euler-angles; m_i is mass; g is gravitational acceleration; c_1, c_2 and l denote the lift coefficient, drag coefficient and the distance from the center of rotor rotation to the mass center, respectively; I_{xx}, I_{yy} and I_{zz} are the inertia tensor diagonal matrices. $\omega_{i,s1}, \omega_{i,s2}, \omega_{i,s3}$ and $\omega_{i,s4}$ are determined by the angular speed of the four motors $\omega_{i,1}, \omega_{i,2}, \omega_{i,3}$ and $\omega_{i,4}$:

$$\begin{cases} \omega_{i,s1} = \omega_{i,1}^2 + \omega_{i,2}^2 + \omega_{i,3}^2 + \omega_{i,4}^2, \\ \omega_{i,s2} = \omega_{i,1}^2 - \omega_{i,3}^2, \\ \omega_{i,s3} = \omega_{i,2}^2 - \omega_{i,4}^2, \\ \omega_{i,s4} = \omega_{i,1}^2 - \omega_{i,2}^2 + \omega_{i,3}^2 - \omega_{i,4}^2. \end{cases} \quad (5)$$

The dynamics model in (5) can be converted to a compact form (7) for analysis.

$$\ddot{Q}_i = \frac{F_i}{m_i} - g[0, 0, 1]^T, \quad (6a)$$

$$\ddot{\Phi}_i = N_i J_i^{-1} - \mathbb{C}_i \dot{\Phi}_i J_i^{-1}, \quad (6b)$$

where $Q_i = [q_{i,x}, q_{i,y}, q_{i,z}]^T \in \mathbb{R}^3$; $\Phi_i = [\phi_i, \theta_i, \psi_i]^T \in \mathbb{R}^3$; $\mathbb{C}_i \in \mathbb{R}^{3 \times 3}$ denotes the vector of Coriolis and centripetal force; $F_i = R[0, 0, T_i]^T \in \mathbb{R}^3$; $J_i \in \mathbb{R}^{3 \times 3}$ is the inertia matrix; $N_i = [N_{\phi_i}, N_{\theta_i}, N_{\psi_i}]^T \in \mathbb{R}^3$, $T_i, N_{\phi_i}, N_{\theta_i}, N_{\psi_i}$ are trust, roll torque,

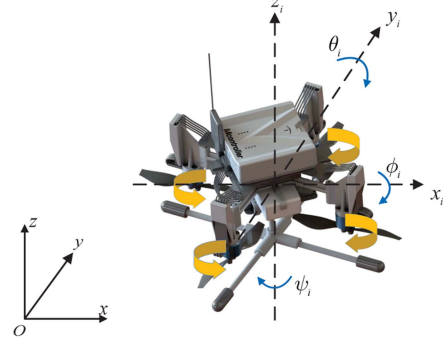


Fig. 1. The system parameters of the quadrotor AAV.

pitch torque and yaw torque respectively; and R_i is shown below:

$$R_i = \begin{bmatrix} c\theta_i c\psi_i & s\phi_i s\theta_i c\psi_i - c\phi_i s\psi_i & c\phi_i s\theta_i c\psi_i + s\phi_i s\psi_i \\ c\theta_i s\psi_i & s\phi_i s\theta_i s\psi_i + c\phi_i c\psi_i & c\phi_i s\theta_i s\psi_i - s\phi_i c\psi_i \\ -s\theta_i & s\phi_i c\theta_i & c\phi_i c\theta_i \end{bmatrix}.$$

The quadrotor AAV parameters are illustrated in Fig. 1, and its dynamics can be transformed into Euler-Lagrange form

$$M_i(q_i) \ddot{q}_i + C_i(q_i, \dot{q}_i) \dot{q}_i + g_i(q_i) = \tau_i, \quad (7)$$

where

$$M_i(q_i) = \begin{bmatrix} m_i I_3 & 0_{3 \times 3} \\ 0_{3 \times 3} & J_i \end{bmatrix}, C_i(q_i, \dot{q}_i) = \begin{bmatrix} 0_{3 \times 3} & 0_{3 \times 3} \\ 0_{3 \times 3} & \mathbb{C}_i \end{bmatrix},$$

$$g_i(q_i) = [0, 0, m_i g, 0, 0, 0]^T, q_i = [Q_i; \Phi_i], \tau_i = [F_i, N_i]^T.$$

Property 1: $M_i(q_i)$ is symmetric positive-definite, and $\dot{M}_i(q_i) - 2C_i(q_i, \dot{q}_i)$ is skew symmetric.

Property 2: For any vector $\hat{q}_i \in \mathbb{R}^6, \dot{\hat{q}}_i \in \mathbb{R}^6$, it has $M_i(q_i) \dot{\hat{q}}_i + C_i(q_i, \dot{\hat{q}}_i) \hat{q}_i + g_i(q_i) = \mathbb{Y}_i(q_i, \dot{\hat{q}}_i, \hat{q}_i) \vartheta_i$, where $\mathbb{Y}_i(q_i, \dot{\hat{q}}_i, \hat{q}_i) \vartheta_i$ is the regressor matrix and $\vartheta_i \in \mathbb{R}^4$ is the integration vector with regard to the system parameters.

Property 3: System terms are all bounded, and there exist positive constants $\mathbb{I}_M, \mathbb{I}_C$ and $\mathbb{I}_g$, such that $\|M_i(q_i)\| \leq \mathbb{I}_M, \|C_i(q_i, \dot{q}_i)\| \leq \mathbb{I}_C \|\dot{q}_i\|$ and $\|g_i(q_i)\| \leq \mathbb{I}_g$.

D. FDI and Physical Attacks

During the operation of quadrotor AAVs, both the cyber and physical systems are vulnerable to various forms of attacks. In this article, cyber attacks primarily involve FDI affecting the NE seeking process and communication among quadrotor AAVs. Physical system attacks primarily focus on disrupting actuators to prevent them from achieving optimal operational states. For instance, as depicted in Fig. 2, consider a system comprising six quadrotor AAVs, in which the communication between quadrotor AAV-4 and quadrotor AAV-5 is subjected to FDI attacks. Additionally, quadrotor AAV-1 experiences FDI attacks in the game layer and physical attacks in the execution layer. In this case, the AAVs achieve their individual goals by minimizing their respective cost functions, thereby collectively accomplishing the global goal of the distributed system.

Inspired by the consensus protocol [29], the noncooperative game model for a second-order system under FDI attacks can

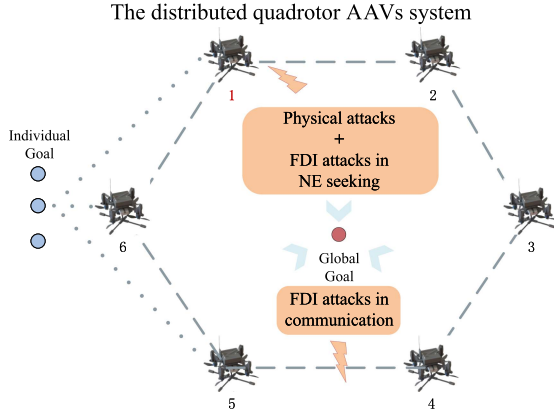


Fig. 2. FDI and physical attacks in a quadrotor AAVs system.

be established as

$$\dot{x}_i = v_i, \quad (8a)$$

$$\dot{v}_i = u_{Gi}, \quad (8b)$$

$$u_{Gi} = -k_1 v_i - \nabla_{x_i} \mathbb{J}_i(x_i, \mathbf{y}_{-i}^i) + \lambda_{G,i}^1 g_{1,i}, \quad (8c)$$

$$\dot{\mathbf{y}}_{-i}^i = -k_2 S_i \sum_{j=1}^n a_{ij} (\mathbf{y}_i - \mathbf{y}_j - \lambda_{G,i}^2 g_{2,j}), \quad (8d)$$

where k_1 and k_2 are positive constants, $\lambda_{G,i}^1$ and $\lambda_{G,i}^2$ are used to determine the presence of FDI attacks. If their values are 1, it indicates the presence of FDI attacks; if their values are 0, it signifies the absence of FDI attacks. $g_{1,i}$ denotes that the agent- i has been subjected to FDI attacks during the decision-making process. $g_{2,j}$ denotes that the communication process between agent- i and agent- j has been subjected to FDI attacks. $x_i \in \mathbb{R}^6$ is the estimate of quadrotor AAV- i for the NE; $\nabla_{x_i} \mathbb{J}_i(x_i, \mathbf{y}_{-i}^i)$ is pseudo-gradient of cost function based on the estimation $\mathbf{y}_{-i}^i$. $\mathbf{y}_{-i}^i = \text{col}\{\mathbf{y}_1^i, \dots, \mathbf{y}_{i-1}^i, \mathbf{y}_{i+1}^i, \dots, \mathbf{y}_n^i\}$ denotes quadrotor AAV- i 's estimate of others. $\mathbf{y}_i = \text{col}\{\mathbf{y}_1^i, \dots, \mathbf{y}_n^i\}$ and $y_i^i = x_i$. S_i is a selection matrix which is defined as

$$S_i = \begin{bmatrix} I_{i-1} & 0_{i-1} & 0_{(i-1) \times (n-i)} \\ 0_{(n-i) \times (i-1)} & 0_{n-i} & I_{n-i} \end{bmatrix} \otimes I_6.$$

According to the properties of the selection matrix S_i [30]. We have $S^T S + R^T R = I$, where $S = \text{blk}\{S_1, \dots, S_n\}$, $R = \text{blk}\{R_1, \dots, R_n\}$, $R_i = [0_{i-1}, 1, 0_{n-i}] \otimes I_6$.

Assumption 6: The FDI attacks $g_{1,i}$ and $g_{2,j}$ and their derivatives with respect to time t are unknown and bounded.

Moreover, considering that the actuators of quadrotor AAVs may be vulnerable to physical attacks and disturbances, which could result in performance degradation, they may fail to attain the desired control input state. The dynamic system of each quadrotor AAV can be transformed into

$$M_i(q_i) \ddot{q}_i + C_i(q_i, \dot{q}_i) \dot{q}_i + g_i(q_i) = \mathcal{G}(\tau_i) + \tau_{di}. \quad (9)$$

where τ_{di} denotes external bounded disturbance, τ_i is the desired control input of the system. $\mathcal{G}(\tau_i) = \rho_i \odot \tau_i + h_{di}$ denotes the actual control input of the system under physical attacks. $\rho_i \in \mathbb{R}^6$ and $h_{di} \in \mathbb{R}^6$ are attack factor and deviation, respectively. If

TABLE I
NOTATION

Symbol	Description
q_i	State vector of AAV i
$\mathbb{J}_i$	Cost function of AAV i
$\nabla_{x_i} \mathbb{J}_i(x_i, \mathbf{y}_{-i}^i)$	Pseudo-gradient of the cost function for AAV i
$\mathcal{L}$	Laplacian matrix representing the communication topology
$g_{1,i}$	Communication FDI attack vector on AAV i
$g_{2,i}$	Computational FDI attack vector on AAV i
ρ_i	Attack factor representing the health of AAV i 's actuator
τ_i	Desired control input of AAV i
τ_{di}	External disturbance input to AAV i

$\rho_i = 1$, the actuators are functional; if not, actuators are in an unhealthy state due to being subjected to physical attacks.

Assumption 7: After the actuators undergo physical attacks, the resulting fault deviation is bounded. $(\tau_{di} + h_{di}) \in \mathcal{L}_\infty$, and note that $D_i = \tau_{di} + h_{di}$, $\|D_{i,k}\| \in (0, \mu_1)$, $\exists \mu_1 > 0$, $\forall i \in \{1, \dots, n\}$, $\forall k \in \{1, \dots, 6\}$.

Assumption 8: The attack factor $\rho_{i,k}$ satisfies $\rho_{i,k} \in [\mu_2, 1)$, $\exists \mu_2 \in (0, 1)$, $\forall i \in \{1, \dots, n\}$, $\forall k \in \{1, \dots, 6\}$.

To facilitate subsequent analysis, we provide Table I to explain the key variables in the system model.

III. MAIN RESULT

In this section, we develop a distributed NE seeking algorithm for quadrotor AAVs operating under both FDI and physical attacks and analyze the robustness of the proposed algorithm. The algorithm comprises two main components: the game layer and the execution layer.

In the game layer, the following algorithm is designed to fulfill the NE seeking under FDI attacks.

$$\dot{x}_i = v_i, \quad (10a)$$

$$\dot{v}_i = u_{Gi}, \quad (10b)$$

$$u_{Gi} = -k_1 \hat{h}_{v,i} - \nabla_{x_i} \mathbb{J}_i(x_i, \mathbf{y}_{-i}^i) + \partial_{1,i} - \hat{\partial}_{1,i}, \quad (10c)$$

$$\dot{\mathbf{y}}_i = -k_2 \sum_{j=1}^n a_{ij} (\mathbf{y}_i - \mathbf{y}_j) + S_i^{-1} \partial_{2,i} - S_i^{-1} \hat{\partial}_{2,i}, \quad (10d)$$

$$\dot{\hat{h}}_{x,i} = \hat{h}_{v,i} + \alpha_1 \mathcal{F}_1(x_i - \hat{h}_{x,i}), \quad (10e)$$

$$\dot{\hat{h}}_{v,i} = -k_1 \hat{h}_{v,i} - \nabla_{x_i} \mathbb{J}_i(x_i, \mathbf{y}_{-i}^i) + \alpha_2 \mathcal{F}_2(x_i - \hat{h}_{x,i}), \quad (10f)$$

$$\dot{\hat{\partial}}_{1,i} = \alpha_2 \mathcal{F}_3(x_i - \hat{h}_{x,i}), \quad (10g)$$

$$\dot{\hat{h}}_{y,i} = -k_2 \sum_{j=1}^n a_{ij} (\mathbf{y}_i - \mathbf{y}_j) + \alpha_4 \mathcal{F}_4(\mathbf{y}_i - \hat{h}_{y,i}), \quad (10h)$$

$$\dot{\hat{\partial}}_{2,i} = \alpha_5 \mathcal{F}_5(\mathbf{y}_i - \hat{h}_{y,i}), \quad (10i)$$

where $\hat{h}_{x,i}$, $\hat{h}_{v,i}$, $\hat{h}_{y,i}$ are the estimations of x_i , v_i and $\mathbf{y}_i$, respectively. Note that $\partial_{1,i} = \lambda_{G,i}^1 g_{1,i}$ and $\partial_{2,i} = k_2 S_i \sum_{j=1}^n a_{ij} \lambda_{G,i}^2 g_{2,j}$. Under Assumption 4, we have $|\hat{\partial}_{1,i}| \leq b_1$,

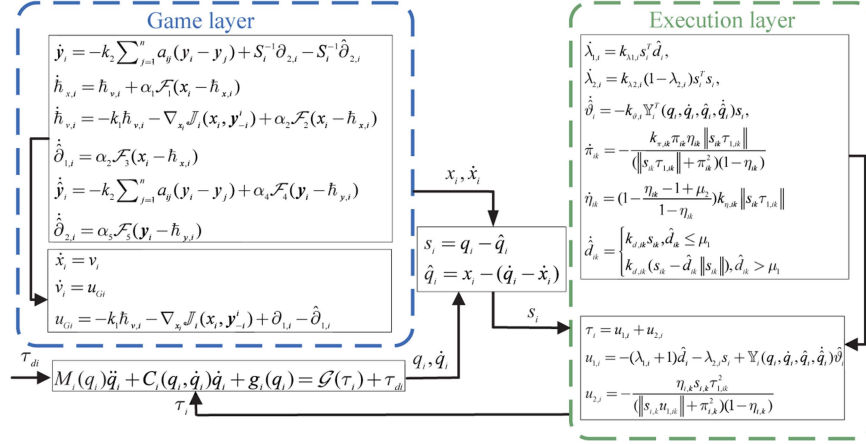


Fig. 3. The framework of the proposed algorithm.

$|\partial_{2,i}| \leq b_2$. $\hat{\partial}_{1,i}$ and $\hat{\partial}_{2,i}$ are the observers of $\partial_{1,i}$ and $\partial_{2,i}$. α_{k_m} ($k_m = 1, 2, 3, 4, 5$) are positive constants. $\mathcal{F}_{k_m}$ are nonlinear functions to be discussed.

In the execution layer, we design the following algorithm with $s_i = q_i - \hat{q}_i$ and $\hat{q}_i = x_i - (\dot{q}_i - \dot{x}_i)$ for quadrotor AAVs to track the NE and address the actuator failure resulting from physical attacks.

$$\tau_i = u_{1,i} + u_{2,i}, \quad (11a)$$

$$u_{1,i} = -(\lambda_{1,i} + 1)\hat{d}_i - \lambda_{2,i}s_i + \mathbb{Y}_i(q_i, \dot{q}_i, \hat{q}_i, \dot{\hat{q}}_i)\hat{\vartheta}_i, \quad (11b)$$

$$u_{2,i} = -\frac{\eta_{ik}s_{ik}\tau_{1,ik}^2}{(\|s_{ik}u_{1,ik}\| + \pi_{ik}^2)(1 - \eta_{ik})}, \quad (11c)$$

where τ_i is a practical controller comprising $u_{1,i}$ and $u_{2,i}$. $u_{1,i}$ is used to handle the unknown dynamics and external disturbances. $u_{2,i}$ is employed to ensure that the system can continue to operate normally even in the event of actuator failures caused by physical attacks. The design of the adaptive laws for the relevant variables is as follows:

$$\dot{\lambda}_{1,i} = k_{\lambda 1,i}s_i^T \hat{d}_i, \quad (12a)$$

$$\dot{\lambda}_{2,i} = k_{\lambda 2,i}(1 - \lambda_{2,i})s_i^T s_i, \quad (12b)$$

$$\dot{\vartheta}_i = -k_{\vartheta,i}\mathbb{Y}_i^T(q_i, \dot{q}_i, \hat{q}_i, \dot{\hat{q}}_i)s_i, \quad (12c)$$

$$\dot{\pi}_{ik} = -\frac{k_{\pi,ik}\pi_{ik}\eta_{ik}\|s_{ik}u_{1,ik}\|}{(\|s_{ik}u_{1,ik}\| + \pi_{ik}^2)(1 - \eta_{ik})}, \quad (12d)$$

$$\dot{\eta}_{ik} = \left(1 - \frac{\eta_{ik} - 1 + \mu_2}{1 - \eta_{ik}}\right)k_{\eta,ik}\|s_{ik}\tau_{1,ik}\|, \quad (12e)$$

$$\dot{d}_{ik} = \begin{cases} k_{d,ik}s_{ik}, & \hat{d}_{ik} \leq \mu_1, \\ k_{d,ik}(s_{ik} - \hat{d}_{ik}\|s_{ik}\|), & \hat{d}_{ik} > \mu_1. \end{cases} \quad (12f)$$

where $k_{\lambda 1,i}, k_{\lambda 2,i}, k_{\pi,ik}, k_{\eta,ik}, k_{d,ik} \in \mathbb{R}$ are positive constants and $k_{\vartheta,i} \in \mathbb{R}^4$ is a positive diagonal matrix. Note that $\eta_{ik} \in [0, 1)$.

Fig. 3 depicts the framework of the algorithm (10), (11) and (12) for the whole distributed quadrotor AAVs system.

Then, the stability of the observers $\hat{\partial}_{1,i}$ and $\hat{\partial}_{2,i}$ need to be analyze. Note that $e_{x,i} = x_i - \hat{x}_{x,i}$, $e_{v,i} =$

$(v_i - \hat{h}_{v,i})/\Theta_0$, $e_{\partial 1,i} = (\partial_{1,i} - \hat{\partial}_{1,i})/\Theta_0^2$, $e_{y,i} = S_i(y_i - \hat{h}_{y,i})$ and $e_{\partial 2,i} = (\partial_{2,i} - \hat{\partial}_{2,i})/\Theta_0$. $e_i = \text{col}\{e_{x,i}, e_{v,i}, e_{\partial 1,i}\}$, $e_{2,i} = \text{col}\{e_{y,i}, e_{\partial 2,i}\}$. Θ_0 is a positive constant.

Assumption 9: There exists positive-definite, continuously differentiable functions V_{o1}, V_{o2}, V_{o3} and V_{o4} along with a series of positive constants $\Lambda_{1,k_n}, \Lambda_{2,k_n}$ ($n = 1, 2, 3, 4$) and $\Gamma_1, \Gamma_2, \Gamma_3$ satisfying the following conditions:

$$\Lambda_{1,k_1}\|e_{1,i}\|^2 \geq V_{o1}(e_{1,i}) \geq \Lambda_{1,k_2}\|e_{1,i}\|^2, \quad (13a)$$

$$\Lambda_{1,k_3}\|e_{1,i}\|^2 \geq V_{o2}(e_{1,i}) \geq \Lambda_{1,k_4}\|e_{1,i}\|^2, \quad (13b)$$

$$\Lambda_{2,k_1}\|e_{2,i}\|^2 \geq V_{o3}(e_{2,i}) \geq \Lambda_{2,k_2}\|e_{2,i}\|^2, \quad (13c)$$

$$\Lambda_{2,k_3}\|e_{2,i}\|^2 \geq V_{o4}(e_{2,i}) \geq \Lambda_{2,k_4}\|e_{2,i}\|^2. \quad (13d)$$

$$\begin{aligned} -V_{o2}(e_{1,i}) &\geq \frac{\partial V_{o1}}{\partial e_{x,i}}(e_{v,i} - \mathcal{F}_1(e_{x,i})) \\ &+ \frac{\partial V_{o1}}{\partial e_{v,i}}(e_{\partial 1,i} - \mathcal{F}_2(e_{x,i})) - \frac{\partial V_{o1}}{\partial e_{\partial 1,i}}\mathcal{F}_3(e_{x,i}). \end{aligned} \quad (14)$$

$$\begin{aligned} -V_{o4}(e_{2,i}) &\geq \frac{\partial V_{o3}}{\partial e_{y,i}}(e_{\partial 2,i} - \mathcal{F}_4(e_{y,i})) - \frac{\partial V_{o3}}{\partial e_{\partial 2,i}}\mathcal{F}_5(e_{y,i}). \end{aligned} \quad (15)$$

$$\Gamma_1\|e_{1,i}\| \geq \left\|\frac{\partial V_{o1}}{\partial e_{\partial 1,i}}\right\|, \quad (16a)$$

$$\Gamma_2\|e_{2,i}\| \geq \left\|\frac{\partial V_{o3}}{\partial e_{y,i}}\right\|, \quad (16b)$$

$$\Gamma_3\|e_{2,i}\| \geq \left\|\frac{\partial V_{o3}}{\partial e_{\partial 2,i}}\right\|. \quad (16c)$$

Assumption 10: $\hat{G}(x)$ represents the estimated value of $G(x)$ and $\hat{G}(x) = \text{col}\{\nabla_{x1}\mathbb{J}_1(x_1, \mathbf{y}_1^1), \dots, \nabla_{xn}\mathbb{J}_n(x_n, \mathbf{y}_n^n)\}$. The map $\hat{G}(x)$ is Lipschitz continuous and satisfies $\|\hat{G}(x_1) - \hat{G}(x_2)\| \leq \varpi\|x_1 - x_2\|$, $\varpi > 0$.

Theorem 1: If Assumptions 6 and 9 hold, $\alpha_1 = \alpha_4 = \Theta_0$, $\alpha_2 = \alpha_5 = \Theta_0^2$ and $\alpha_3 = \Theta_0^3$, the observer 10(e)–10(i) exhibit high-order infinitesimal observation errors as $\Theta_0 \rightarrow 0$. It is $\lim_{t \rightarrow \infty} \sup |e_{1,i}| \leq [\mathcal{O}(\frac{1}{\Theta_0^3}), \mathcal{O}(\frac{1}{\Theta_0^2}), \mathcal{O}(\frac{1}{\Theta_0})]^T$ and $\lim_{t \rightarrow \infty} \sup |e_{2,i}| \leq [\mathcal{O}(\frac{1}{\Theta_0}), \mathcal{O}(\frac{1}{\Theta_0})]^T$.

Proof: For observer systems 10(e)–10(g), based on the errors $e_{1,i}$, its derivative can be written as:

$$\dot{e}_{1,i} = \Theta_0 \begin{bmatrix} e_{v,i} - \mathcal{F}_1(e_{x,i}) \\ e_{\partial_1,i} - \mathcal{F}_2(e_{x,i}) \\ \frac{\dot{\partial}_{1,i}}{\Theta_0^3} - \mathcal{F}_3(e_{x,i}) \end{bmatrix}. \quad (17)$$

The partial derivative of V_{o1} with respect to $e_{1,i}$ is

$$\begin{aligned} \dot{V}_{o1}(e_{1,i}) &= \Theta_0 \frac{\partial V_{o1}}{\partial e_{x,i}} (e_{v,i} - \mathcal{F}_1(e_{x,i})) \\ &+ \Theta_0 \frac{\partial V_{o1}}{\partial e_{v,i}} (e_{\partial_1,i} - \mathcal{F}_2(e_{x,i})) \\ &- \Theta_0 \frac{\partial V_{o1}}{\partial e_{\partial_1,i}} \mathcal{F}_3(e_{x,i}) + \frac{\partial V_{o1}}{\partial e_{\partial_1,i}} \frac{\dot{\partial}_{1,i}}{\Theta_0^2}. \end{aligned} \quad (18)$$

Based on (14) in Assumption 9, it gives that

$$\dot{V}_{o1}(e_{1,i}) \leq -\Theta_0 V_{o2}(e_{1,i}) + \frac{\partial V_{o1}}{\partial e_{\partial_1,i}} \frac{\dot{\partial}_{1,i}}{\Theta_0^2}. \quad (19)$$

Then, by Assumption 6 and Assumption 9, it yields

$$\dot{V}_{o1} \leq -\Theta_0 \frac{\Lambda_{1,k_3}}{\Lambda_{1,k_2}} V_{o1}(e_{1,i}) + \frac{\Gamma_1 b_1 \sqrt{\Lambda_{1,k_1}}}{\Theta_0^2 \Lambda_{1,k_1}} \sqrt{V_{o1}(e_{1,i})}. \quad (20)$$

Then, it gives that

$$\begin{aligned} \|e_{1,i}\| &\leq \frac{\sqrt{\Lambda_{1,k_1}}}{\Lambda_{1,k_1}} \sqrt{V_{o1}(t)} \\ &\leq \frac{\sqrt{\Lambda_{1,k_1}}}{\Lambda_{1,k_1}} \sqrt{V_{o1}(0)} e^{-\frac{\Theta_0 \Lambda_{1,k_3}}{2 \Lambda_{1,k_2}} t} \\ &+ \Gamma_1 b_1 \frac{\Lambda_{1,k_2}}{\Lambda_{1,k_1} \Lambda_{1,k_3} \Theta_0^3} \left(1 - e^{-\frac{\Theta_0 \Lambda_{1,k_3}}{2 \Lambda_{1,k_2}} t}\right). \end{aligned} \quad (21)$$

Thus, we can obtain that

$$\lim_{t \rightarrow \infty} \sup |e_{1,i}| \leq \left[\mathcal{O}\left(\frac{1}{\Theta_0^3}\right), \mathcal{O}\left(\frac{1}{\Theta_0^2}\right), \mathcal{O}\left(\frac{1}{\Theta_0}\right) \right]^T. \quad (22)$$

For observer 10(h)–10(i), we have

$$\dot{e}_{2,i} = \Theta_0 \begin{bmatrix} e_{\partial_2,i} - \mathcal{F}_4(e_{y,i}) \\ \frac{\dot{\partial}_{2,i}}{\Theta_0^3} - \mathcal{F}_5(e_{y,i}) \end{bmatrix}. \quad (23)$$

The partial derivative of V_{o3} with respect to $e_{2,i}$ is

$$\begin{aligned} \dot{V}_{o3}(e_{2,i}) &= \Theta_0 \frac{\partial V_{o3}}{\partial e_{y,i}} (e_{\partial_2,i} - \mathcal{F}_4(e_{y,i})) - \Theta_0 \frac{\partial V_{o3}}{\partial e_{\partial_2,i}} \\ &\times \mathcal{F}_5(e_{y,i}) + \frac{\partial V_{o3}}{\partial e_{\partial_2,i}} \frac{\dot{\partial}_{2,i}}{\Theta_0}. \end{aligned} \quad (24)$$

Based on (15) in Assumption 9, it gives that

$$\dot{V}_{o3}(e_{2,i}) \leq -\Theta_0 V_{o4}(e_{2,i}) + \frac{\partial V_{o3}}{\partial e_{\partial_2,i}} \frac{\dot{\partial}_{2,i}}{\Theta_0}. \quad (25)$$

By Assumption 6 and Assumption 9, it obtains that

$$\begin{aligned} \dot{V}_{o3} &\leq -\Theta_0 \frac{\Lambda_{2,k_3}}{\Lambda_{2,k_2}} V_{o3}(e_{2,i}) \\ &+ \frac{\Gamma_3 b_2 \sqrt{\Lambda_{2,k_1}}}{\Theta_0 \Lambda_{2,k_1}} \sqrt{V_{o3}(e_{2,i})}. \end{aligned} \quad (26)$$

Then, the analysis process is similar to (21). We have

$$\begin{aligned} \|e_{2,i}\| &\leq \frac{\sqrt{\Lambda_{2,k_1}}}{\Lambda_{2,k_1}} \sqrt{V_{o3}(0)} e^{-\frac{\Theta_0 \Lambda_{2,k_3}}{2 \Lambda_{2,k_2}} t} \\ &+ \Gamma_3 b_2 \frac{\Lambda_{2,k_2}}{\Lambda_{2,k_1} \Lambda_{2,k_3} \Theta_0} \left(1 - e^{-\frac{\Theta_0 \Lambda_{2,k_3}}{2 \Lambda_{2,k_2}} t}\right). \end{aligned} \quad (27)$$

Thus, it derives that

$$\lim_{t \rightarrow \infty} \sup |e_{2,i}| \leq \left[\mathcal{O}\left(\frac{1}{\Theta_0}\right), \mathcal{O}\left(\frac{1}{\Theta_0}\right) \right]^T. \quad (28)$$

Finally, the proof of Theorem 1 is completed. $\square$

The compact form of the system without attacks is

$$\dot{x} = v, \quad (29a)$$

$$\dot{v} = u_G, \quad (29b)$$

$$u_G = -k_1 v - \hat{G}(\mathbf{y}), \quad (29c)$$

$$\dot{y} = -k_2 \mathbf{L} \mathbf{y}, \quad (29d)$$

where $x = \text{col}\{x_1, \dots, x_n\}$, $v = \text{col}\{v_1, \dots, v_n\}$, $u_G = \text{col}\{u_{G1}, \dots, u_{Gn}\}$, $\mathbf{y} = \text{col}\{\mathbf{y}_1, \dots, \mathbf{y}_n\}$, $\hat{G}(x) = \text{col}\{\nabla_{x1} \mathbb{J}_1(x_1, \mathbf{y}_{-1}^1), \dots, \nabla_{xn} \mathbb{J}_n(x_n, \mathbf{y}_{-n}^n)\}$ and $\mathbf{L} = L \otimes I_n$.

Theorem 2: Suppose that Assumptions 1, 4, 5 and 10 hold. $x^* \in \mathbb{R}^{6n}$ is a NE of the noncooperative game if and only if there exist $v^* = 0_{6n}$ and $\mathbf{y}^* = 1_n \otimes x^*$ such that $(x^*, v^*, \mathbf{y}^*)$ is the NE of the system (29).

Proof: (I) According to the NE of the system (29), we have

$$0_{6n} = v^*, \quad (30a)$$

$$0_{6n} = -k_1 v^* - \hat{G}(\mathbf{y}^*), \quad (30b)$$

$$0_{36n^2} = -k_2 \mathbf{L} \mathbf{y}^*. \quad (30c)$$

It's obvious that $v^* = 0_{6n}$ and $G(\mathbf{y}^*) = \hat{G}(\mathbf{y}^*) = 0_{6n}$. The communication graph $\mathbb{G}$ is undirected and connected. It gives $\mathbf{y}^* = 1_n \otimes x^*$. Thus, according to Lemma 1, x^* is a NE of the noncooperative game.

(II) Suppose that x^* is a NE of the noncooperative game. We have $G(\mathbf{y}^*) = \hat{G}(\mathbf{y}^*) = 0_{6n}$ based on Lemma 1 and Assumption 10. It gives that $\mathbf{L} \mathbf{y}^* = 0_{36n^2}$ based on undirected and connected graph $\mathbb{G}$. It follows from (29) that $(x^*, 0_{6n}, \mathbf{y}^*)$ is the NE with $v^* = 0_{6n}$.

Theorem 3: The distributed quadrotor AAVs system under strategies (10) can asymptotically seek a narrow neighborhood of the NE of the noncooperative game based on Assumptions 1–6, 9 and 10.

Proof: Let $S = \text{blk}\{S_1, \dots, S_n\}$, $S \mathbf{y} = \text{col}\{\mathbf{y}_{-1}^1, \dots, \mathbf{y}_{-n}^n\}$, $e_v = \text{col}\{e_{v,1}, \dots, e_{v,n}\}$, $e_{\partial_1} = \text{col}\{e_{\partial_1,1}, \dots, e_{\partial_1,n}\}$ and $e_{\partial_2} = \text{col}\{e_{\partial_2,1}, \dots, e_{\partial_2,n}\}$. The compact form of (8) is

$$\dot{x} = v, \quad (31a)$$

$$\dot{v} = u_G, \quad (31b)$$

$$u_G = -k_1 v - \hat{G}(\mathbf{y}) + \Theta_0(e_v + \Theta_0 e_{\partial_1}), \quad (31c)$$

$$\dot{\mathbf{y}} = -k_2 L \mathbf{y} + \Theta_0 S^{-1} e_{\partial_2}. \quad (31d)$$

Construct the candidate Lyapunov function

$$\begin{aligned} V_1 &= \frac{1}{2} \|v - v^*\|^2 + \frac{1}{2} \|k_1(x - x^*) + (v - v^*)\|^2 \\ &\quad + \frac{1}{2} (\mathbf{y} - \mathbf{y}^*)^T S^T S (\mathbf{y} - \mathbf{y}^*). \\ \dot{V}_1 &= -k_1 \|v - v^*\|^2 - 2(v - v^*)^T (\hat{G}(\mathbf{y}) - \hat{G}(\mathbf{y}^*)) \\ &\quad - k_1(x - x^*)^T (\hat{G}(\mathbf{y}) - \hat{G}(\mathbf{y}^*)) \\ &\quad - k_2 (\mathbf{y} - \mathbf{y}^*)^T S^T S \mathbf{L} (\mathbf{y} - \mathbf{y}^*) \\ &\quad + k_1(x - x^*)^T (\Theta_0 e_v + \Theta_0^2 e_{\partial_1}) \\ &\quad + \Theta_0 (\mathbf{y} - \mathbf{y}^*)^T S^T e_{\partial_2} \\ &\quad + 2(v - v^*)^T (\Theta_0 e_v + \Theta_0^2 e_{\partial_1}). \end{aligned} \quad (32)$$

Based on Lemma 1, Assumptions 2 and 10, it gives that

$$\begin{aligned} &-2(v - v^*)^T (\hat{G}(\mathbf{y}) - \hat{G}(\mathbf{y}^*)) \\ &= -2(v - v^*)^T (\hat{G}(\mathbf{y}) \\ &\quad - \hat{G}(1_n \otimes x)) - 2(v - v^*)^T (G(x) - G(x^*)) \\ &\leq 2\wp \|v - v^*\| \|x - x^*\| \\ &\quad + 2\varpi \|v - v^*\| \left\| \left(I_{n^2} - \frac{1}{n} 1_n 1_n^T \otimes I_n \right) (\mathbf{y} - \mathbf{y}^*) \right\| \\ &\leq \wp \|x - x^*\|^2 + (\wp + \varpi) \|v - v^*\|^2 \\ &\quad + \varpi \left\| \left(I_{n^2} - \frac{1}{n} 1_n 1_n^T \otimes I_n \right) (\mathbf{y} - \mathbf{y}^*) \right\|^2. \end{aligned} \quad (33)$$

where $\hat{G}(1_n \otimes x^*) = G(x^*) = 0_n$ by the definitions of $\hat{G}(\mathbf{y})$ and $G(x)$.

According to $S^T S + R^T R = I$, we have

$$\begin{aligned} &-k_2 (\mathbf{y} - \mathbf{y}^*)^T S^T S \mathbf{L} (\mathbf{y} - \mathbf{y}^*) \\ &= -k_2 (\mathbf{y} - \mathbf{y}^*)^T \mathbf{L} (\mathbf{y} - \mathbf{y}^*) \\ &\quad + k_2 (\mathbf{y} - \mathbf{y}^*)^T R^T R \mathbf{L} (\mathbf{y} - \mathbf{y}^*) \\ &\leq -k_2 \lambda_2 \left\| \left(I_{n^2} - \frac{1}{n} 1_n 1_n^T \otimes I_n \right) (\mathbf{y} - \mathbf{y}^*) \right\|^2 \\ &\quad + k_2 \|R\mathbf{L}\| \|x - x^*\| \left\| \left(I_{n^2} - \frac{1}{n} 1_n 1_n^T \otimes I_n \right) (\mathbf{y} - \mathbf{y}^*) \right\| \\ &\leq -\left(k_2 \lambda_2 - \frac{k_2}{2} \|R\mathbf{L}\| \right) \left\| \left(I_{n^2} - \frac{1}{n} 1_n 1_n^T \otimes I_n \right) (\mathbf{y} - \mathbf{y}^*) \right\|^2 \\ &\quad + \frac{k_2}{2} \|R\mathbf{L}\| \|x - x^*\|^2. \end{aligned} \quad (34)$$

Furthermore,

$$-k_1(x - x^*)^T (\hat{G}(\mathbf{y}) - \hat{G}(\mathbf{y}^*))$$

$$\begin{aligned} &= -k_1(x - x^*)^T (\hat{G}(\mathbf{y}) - \hat{G}(1_n \otimes x)) \\ &\quad - k_1(x - x^*)^T (G(x) - G(x^*)) \\ &\leq -\frac{k_1 \omega}{2} \|x - x^*\|^2 \\ &\quad + \frac{k_1 \varpi^2}{2\omega} \left\| \left(I_{n^2} - \frac{1}{n} 1_n 1_n^T \otimes I_n \right) (\mathbf{y} - \mathbf{y}^*) \right\|^2. \end{aligned} \quad (35)$$

Therefore, it yields that

$$\begin{aligned} \dot{V}_1 &\leq -\left(\frac{k_1 \omega}{2} - \frac{k_2}{2} \|R\mathbf{L}\| - \wp \right) \|x - x^*\|^2 \\ &\quad - (k_1 - \wp - \varpi) \|v - v^*\|^2 \\ &\quad - \left(k_2 \lambda_2 - \frac{k_2}{2} \|R\mathbf{L}\| - \varpi - \frac{k_1 \varpi^2}{2\omega} \right) \\ &\quad \times \left\| \left(I_{n^2} - \frac{1}{n} 1_n 1_n^T \otimes I_n \right) (\mathbf{y} - \mathbf{y}^*) \right\|^2 \\ &\quad + \Theta_0 \left\| \left(I_{n^2} - \frac{1}{n} 1_n 1_n^T \otimes I_n \right) (\mathbf{y} - \mathbf{y}^*) \right\| \\ &\quad \times \|e_{\partial_2}\| + k_1 \Theta_0 \|x - x^*\| \|e_v + \Theta_0 e_{\partial_1}\| \\ &\quad + 2\Theta_0 \|v - v^*\| \|e_v + \Theta_0 e_{\partial_1}\|. \end{aligned} \quad (36)$$

According to Theorem 1, we have

$$\limsup_{t \rightarrow \infty} \|e_{\partial_2}\| = \mathcal{O}\left(\frac{1}{\Theta_0}\right), \quad (37a)$$

$$\limsup_{t \rightarrow \infty} \|\Theta_0 e_v + \Theta_0^2 e_{\partial_1}\| = \mathcal{O}\left(\frac{1}{\Theta_0}\right). \quad (37b)$$

Substituting (37) into (36) and combining input-to-state stability, it gives that when both

$$k_1 > \max \left\{ \frac{2\wp + k_2 \|R\mathbf{L}\|}{\omega}, \wp + \varpi \right\} \quad (38)$$

and

$$k_2 > \frac{k_1 \varpi^2 + 2\omega \varpi}{2\omega \lambda_2 - \omega \|R\mathbf{L}\|}. \quad (39)$$

are satisfied, we can obtain

$$\limsup_{t \rightarrow \infty} \|x - x^*\| = \mathcal{O}\left(\frac{1}{\Theta_0}\right). \quad (40)$$

It implies that the strategies of all quadrotor AAVs can seek a narrow neighborhood around the NE of the noncooperative game. The proof of Theorem 3 is completed. $\square$

Theorem 4: Even in the event of the quadrotor AAVs undergoing physical attacks resulting in actuator failures, the distributed quadrotor AAVs system under algorithm (11)–(12) can track the narrow neighborhood of NE found by algorithm (10) based on Assumptions 7 and 8.

Proof: Let $\tilde{v}_i = v_i - \hat{v}_i$, $\tilde{\eta}_{ik} = 1 - \eta_{ik} - \mu_2$, and $\tilde{d}_{ik} = D_{ik} - \hat{d}_{ik}$. Substituting algorithms (11)–(12) into dynamic system (9) of each quadrotor AAV, it gives the closed-loop system

$$\dot{\tilde{v}}_i = k_{\vartheta, i} \mathbb{Y}_i^T s_i, \quad (41a)$$

$$\dot{\tilde{\eta}}_{ik} = - \left(1 + \frac{\tilde{\eta}_{ik}}{1 - \eta_{ik}} \right) k_{\eta,ik} \|s_{ik} u_{1,ik}\|, \quad (41b)$$

$$\dot{\tilde{d}}_{ik} = \begin{cases} -\kappa_{d,ik} s_{ik} & \hat{d}_{ik} \leq \mu_1, \\ -\kappa_{d,ik} (s_{ik} - \hat{d}_{ik} \|s_{ik}\|) & \hat{d}_{ik} > \mu_1, \end{cases} \quad (41c)$$

$$\dot{s}_i = M_i^{-1} (D_i + \rho_i \odot \tau_i - C_i s_i - \mathbb{Y}_i \vartheta_i), \quad (41d)$$

where M_i , C_i and $\mathbb{Y}_i$ represent abbreviated expressions for $M_i(q_i)$, $C_i(q_i)$, and $\mathbb{Y}_i(q_i, \dot{q}_i, \hat{q}_i, \dot{\hat{q}}_i)$, respectively. Construct the candidate Lyapunov function.

$$V_2 = \frac{1}{2} s_i^T M_i s_i + \frac{1}{2} \tilde{\vartheta}_i^T k_{\vartheta,i}^{-1} \tilde{\vartheta}_i + \frac{1}{2} k_{\lambda 1,i}^{-1} \lambda_{1,i}^2 + \frac{1}{2} k_{\lambda 2,i}^{-1} \lambda_{2,i}^2 + \frac{1}{2} \sum_{k=1}^6 \left(\mu_2 k_{\pi,ik}^{-1} \pi_{ik}^2 + k_{\eta,ik}^{-1} \tilde{\eta}_{ik}^2 + k_{d,ik}^{-1} \tilde{d}_{ik}^2 \right). \quad (42)$$

Its derivative is

$$\begin{aligned} \dot{V}_2 = & \sum_{k=1}^6 \left(\mu_2 k_{\pi,ik}^{-1} \pi_{ik} \dot{\pi}_{ik} + k_{\eta,ik}^{-1} \tilde{\eta}_{ik} \dot{\tilde{\eta}}_{ik} + k_{d,ik}^{-1} \tilde{d}_{ik} \dot{\tilde{d}}_{ik} \right) \\ & + \frac{1}{2} s_i^T \dot{M}_i s_i + s_i^T M_i \dot{s}_i + \tilde{\vartheta}_i^T k_{\vartheta,i}^{-1} \dot{\tilde{\vartheta}}_i + k_{\lambda 1,i}^{-1} \lambda_{1,i} \dot{\lambda}_{1,i} \\ & + k_{\lambda 2,i}^{-1} \lambda_{2,i} \dot{\lambda}_{2,i}. \end{aligned} \quad (43)$$

Substituting (10)–(11) and (40) into (42), it yielded that

$$\begin{aligned} \dot{V}_2 = & - \sum_{k=1}^6 \frac{\rho_{ik} \eta_{ik} s_{ik}^2 u_{1,ik}^2 + \mu_2 \eta_{ik} \pi_{ik}^2 \|s_{ik} u_{1,ik}\|}{(\|s_{ik} u_{1,ik}\| + \pi_{ik}^2)(1 - \eta_{ik})} \\ & - \sum_{k=1}^6 \left(\tilde{\eta}_{ik}^T - \frac{\mu_2 + \eta_{ik} - 1}{1 - \eta_{ik}} \right) \|s_{ik} u_{1,ik}\| \\ & + \sum_{k=1}^6 \frac{(\rho_{ik} - 1)(1 - \eta_{ik})}{1 - \eta_{ik}} s_{ik} u_{1,ik} + \sum_{k=1}^6 \tilde{d}_{ik}^T k_{d,ik}^{-1} \dot{\tilde{d}}_{ik} \\ & + \lambda_{1,i} s_i^T \hat{d}_i + s_i^T D_i + \lambda_{2,i} (1 - \lambda_{2,i}) s_i^2 \\ & - s_i^T \mathbb{Y}_i \hat{\vartheta}_i + s_i^T u_{1,i}. \end{aligned} \quad (44)$$

Under Assumption 6, we have

$$\begin{aligned} \dot{V}_2 \leq & - \sum_{k=1}^6 \frac{\mu_2 \eta_{ik} - (1 - \mu_2)(1 - \eta_{ik})}{1 - \eta_{ik}} \|s_{ik} u_{1,ik}\| \\ & - \sum_{k=1}^6 \left(\tilde{\eta}_{ik}^T + \frac{\tilde{\eta}_{ik} \tilde{\eta}_{ik}}{1 - \eta_{ik}} \right) \|s_{ik} u_{1,ik}\| \\ & + \sum_{k=1}^6 \tilde{d}_{ik}^T k_{d,ik}^{-1} \dot{\tilde{d}}_{ik} - \lambda_{2,i}^2 s_i^2 + s_i^T \tilde{d}_i. \end{aligned} \quad (45)$$

Then, we proceed with the next step of derivation.

$$\begin{aligned} \dot{V}_2 \leq & - \sum_{k=1}^6 \frac{\tilde{\eta}_{ik}^T \tilde{\eta}_{ik}}{1 - \eta_{ik}} \|s_{ik} u_{1,ik}\| + \sum_{k=1}^6 \tilde{d}_{ik}^T k_{d,ik}^{-1} \dot{\tilde{d}}_{ik} \\ & - \lambda_{2,i}^2 s_i^2 + s_i^T \tilde{d}_i. \end{aligned} \quad (46)$$

TABLE II
INITIAL STATE INFORMATION OF QUADROTOR AAVs

Initial state	$q_1(0)$	$q_2(0)$	$q_3(0)$	$q_4(0)$	$q_5(0)$	$q_6(0)$
$q_{ix}(0)(m)$	3	0	1	0.5	0.5	0.5
$q_{iy}(0)(m)$	-2	0	0.5	0.1	-0.2	-0.2
$q_{iz}(0)(m)$	1	2	0.5	-0.2	0.5	0.4
$q_{i\phi}(0)(rad)$	-1	2.5	1.5	-0.3	-0.6	0.8
$q_{i\theta}(0)(rad)$	-1	-2	1.5	0.2	0.7	-0.8
$q_{i\varphi}(0)(rad)$	1	-2.5	0.6	-0.2	-0.5	-0.5

TABLE III
CHARACTERISTIC PARAMETERS OF QUADROTOR AAVs

Initial state	Q1	Q2	Q3	Q4	Q5	Q6
$m_i(kg)$	2.5	2.5	2.5	2.5	2.5	2.5
$I_{ixx}(kg.m^2)$	0.028	0.031	0.042	0.035	0.04	0.05
$I_{iyy}(kg.m^2)$	0.028	0.031	0.042	0.035	0.04	0.05
$I_{izz}(kg.m^2)$	0.057	0.065	0.07	0.06	0.08	0.095

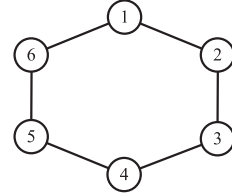


Fig. 4. The normal communication networks of six quadrotor AAVs.

TABLE IV
CONTROL GAIN PARAMETERS

k_1	k_2	Θ_0	μ_2	$\rho_{i,k}$	$k_{\vartheta,i}$
8	10	50	0.5	0.7	10
$k_{d,ik}$	$k_{\lambda 1,i}$	$k_{\lambda 2,i}$	$k_{\pi,ik}$	$k_{\eta,ik}$	
10	10	10	0.1	0.05	

Under the condition that $\eta_{ik} \in [0, 1)$, we have $-\sum_{k=1}^6 \frac{\tilde{\eta}_{ik}^T \tilde{\eta}_{ik}}{1 - \eta_{ik}} \|s_{ik} u_{1,ik}\| \leq 0$.

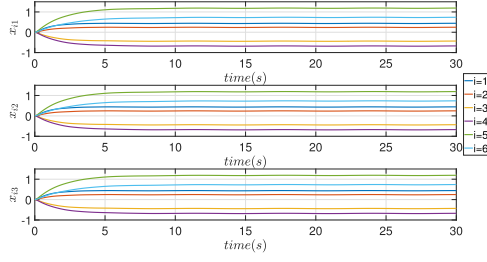
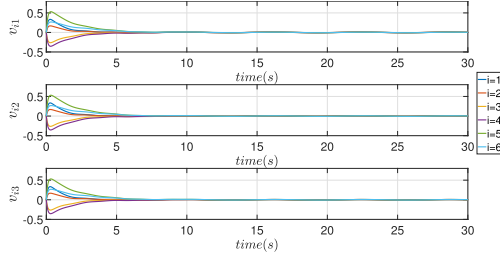
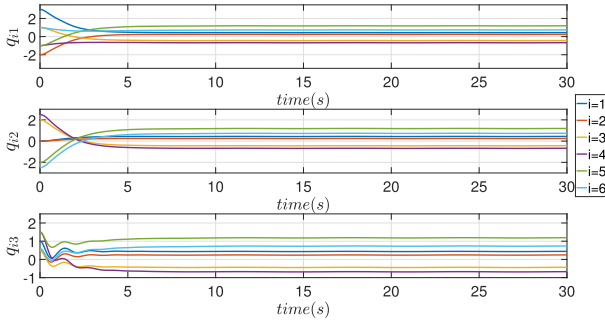
Thus, $\dot{V}_2 \leq \sum_{k=1}^6 \tilde{d}_{ik}^T k_{d,ik}^{-1} \dot{\tilde{d}}_{ik} + s_i^T \tilde{d}_i$, and

$$\dot{V}_2 \leq \begin{cases} 0 & \hat{d}_{ik} \leq \kappa_1, \\ \sum_{k=1}^6 \tilde{d}_{ik}^T \hat{d}_{ik} \|s_{ik}\| & \hat{d}_{ik} > \kappa_1. \end{cases} \quad (47)$$

If $\hat{d}_{ik} > \kappa_1 > 0$ holds, it yields that $\tilde{d}_{ik} < 0$ and $\sum_{k=1}^6 \tilde{d}_{ik}^T \hat{d}_{ik} \|s_{ik}\| \leq 0$.

Finally, we can obtain $\dot{V}_2 \leq 0$. It means that $s_i, \tilde{\vartheta}_i, \lambda_{1,i}, \lambda_{2,i}, \pi_i, \tilde{\eta}_i, \tilde{d}_i$ and $\dot{s}_i \in \mathbb{L}_\infty$. Therefore, it can be concluded that $\lim_{t \rightarrow \infty} \|s_i(t)\| = 0$ by Barbalat's lemma. The distributed quadrotor AAVs system can track the narrow neighborhood of NE under physical attacks. $\square$

Remark 1: In this study, the Lyapunov method is employed to analyze the stability of a specific second-order nonlinear quadrotor AAV system. The Lyapunov framework effectively


 Fig. 5. The estimated positions x_i of six quadrotor AAVs.

 Fig. 6. The estimated velocities v_i of six quadrotor AAVs.

 Fig. 7. The actual positions q_i of six quadrotor AAVs.

demonstrates asymptotic stability under disturbances and attacks. However, it is important to note the following limitations: (1) The stability proof is contingent on the selection of a suitable Lyapunov function, which may limit its generalizability to broader classes of nonlinear systems. (2) Lyapunov-based analysis predominantly ensures local stability, which might not hold when the system experiences significant deviations from the desired equilibrium. To address these concerns, we have incorporated Input-to-State Stability (ISS) analysis and Barbalat's Lemma in our theoretical framework. These tools extend the scope of the stability analysis by establishing a connection between bounded state behaviors and asymptotic convergence. While the current analysis is tailored to the distributed quadrotor AAVs system, we acknowledge the value of exploring global stability tools and frameworks to enhance the generalizability and robustness of the proposed approach in future studies.

Remark 2: This study adopts NE as the control objective for a multi-AAV system, leveraging its well-defined properties within game-theoretic frameworks. Under the constraints of Assumptions 4 and 5, the existence and uniqueness of the NE point can be guaranteed. However, several limitations of NE are acknowledged: (1) In more complex systems with discontinuous

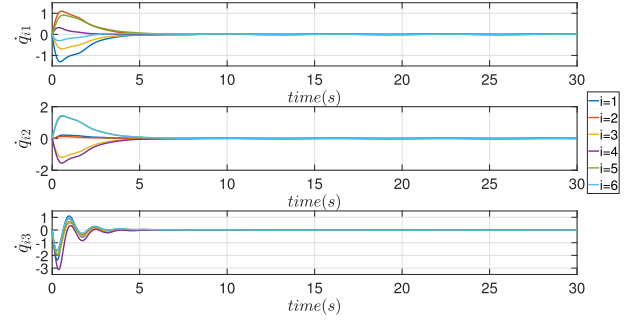
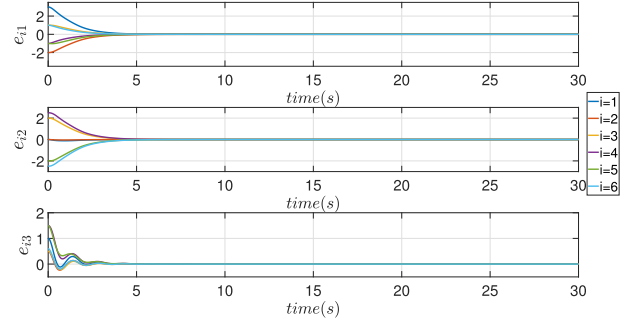
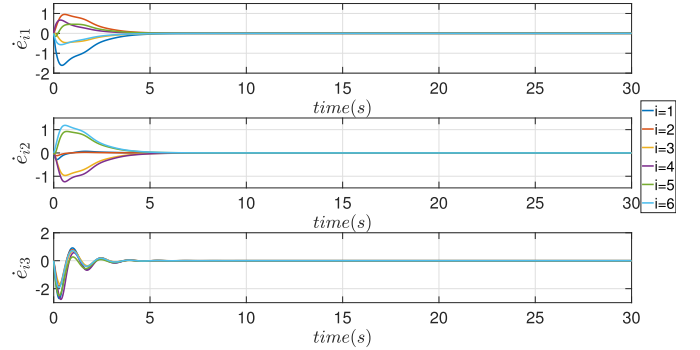
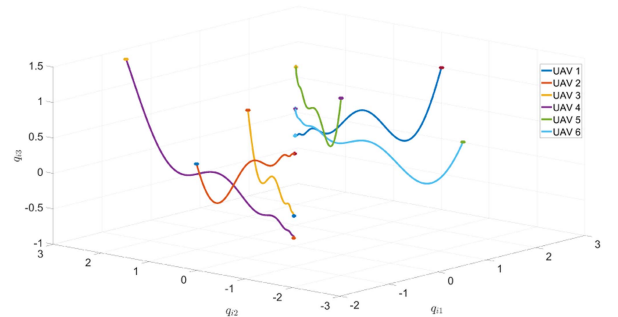

 Fig. 8. The actual velocities $\dot{q}_i$ of six quadrotor AAVs.

 Fig. 9. The positions tracking errors e_i of six quadrotor AAVs.

 Fig. 10. The velocities tracking errors $\dot{e}_i$ of six quadrotor AAVs.


Fig. 11. The trajectories of six quadrotor AAVs.

cost functions, or without Assumption 5, multiple equilibria may arise, some of which could be unstable or suboptimal. (2) In distributed AAV collaborations, inter-agent dependencies are significant and may not fully align with the assumption of independent decision-making. To address these concerns, this paper

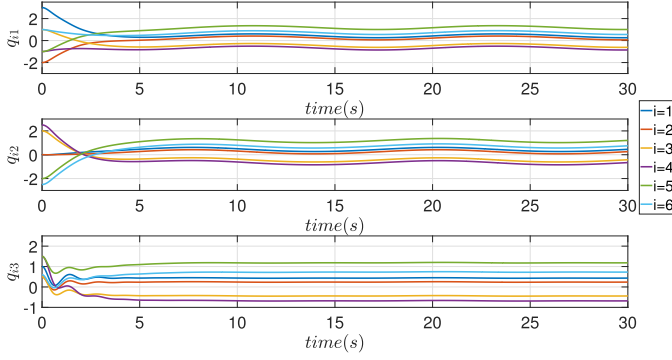


Fig. 12. The actual positions q_i without disturbance rejection term.

validates the uniqueness of the NE through rigorous theoretical analysis and simulation results. Additionally, future research could explore the integration of alternative game-theoretic approaches, such as mixed strategy equilibria and evolutionary stable strategies, to better model the dynamics and uncertainties inherent in AAV systems.

IV. SIMULATION

In this section, a series of simulations are conducted to demonstrate the effectiveness of the algorithm proposed in the preceding sections. A noncooperative game involving six quadrotor AAVs is considered, with the characteristic parameters and initial states of each AAV detailed in Tables II and III. The corresponding communication graph is illustrated in Fig. 4. The AAV model parameters are based on [22], [23], and the cost function for each AAV is defined as: $\mathbb{J}_i(q_i, q_{-i}) = \frac{1}{2}q_i^T(Z_{1,i} \otimes I_6)q_i + q_i^T(Z_{2,i} \otimes I_6) + \frac{1}{3}q_i^T \sum_{i=1}^6 q_i$, where $Z_{1,i}$ and $Z_{2,i}$ are the i th elements of $Z_1 = [9, 8, 7, 6, 5, 4]^T$ and $Z_2 = [-4, -2, 3, 4, -6, -3]^T$, respectively, following similar examples in [37], [40], [41]. The FDI attacks are modeled as $g_{1,i} = \frac{1}{2}[\sin(t), \sin(t), \sin(t), 0, 0, 0]^T$ and $g_{2,j} = \frac{1}{2}\cos(t)I_{36 \times 1}$ with $\lambda_{G,i}^1 = 1$ and $\lambda_{G,i}^2 = 1$ to represent scenarios where both attack types are active. The parameter settings for the control gains are shown in Table IV. The initial values $\pi_{ik} = 0$ and $\eta_{ik} = 0$. The external input disturbances $\tau_{d,i}$ are set to be $[\sin(0.5t), \cos(0.5t), \sin(0.5t), \sin(t), \sin(t), \cos(t)]^T$. In the game layer, the parameters k_1 and k_2 must satisfy the convergence conditions outlined in (39) and (40) to enable the system to compute the NE point. As seen from (41), Θ_0 determines the precision of convergence to the NE point. In the execution layer, $k_{\vartheta,i}$, $k_{d,i}$, $k_{\lambda 1,i}$, $k_{\lambda 2,i}$, $k_{\pi,ik}$ and $k_{\eta,ik}$ are the adaptive gain parameters of virtual variables. Their values primarily affect the convergence speed of the execution layer. Small values will prolong the system's convergence time, while excessively large values can cause significant oscillations in the actual positions and velocities of the AAVs in the execution layer.

The estimated positions x_i and velocities v_i in three-dimensional space in the game layer are illustrated in Figs. 5 and 6. These results indicate that, even in the presence of both types of FDI attacks, all quadrotor AAVs are able to compute positions near the NE points. The actual positions q_i and velocities $\dot{q}_i$ in three-dimensional space in the execution layer are

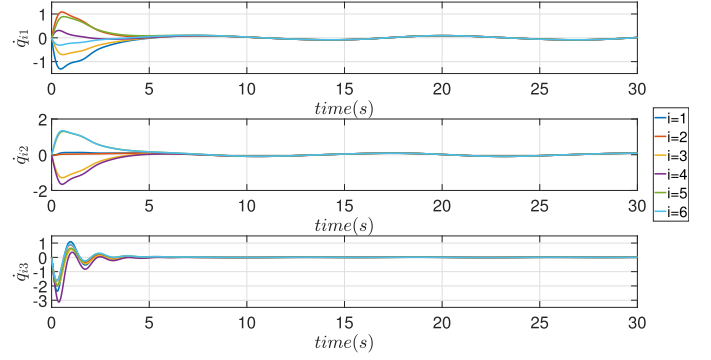


Fig. 13. The actual velocities $\dot{q}_i$ without disturbance rejection term.

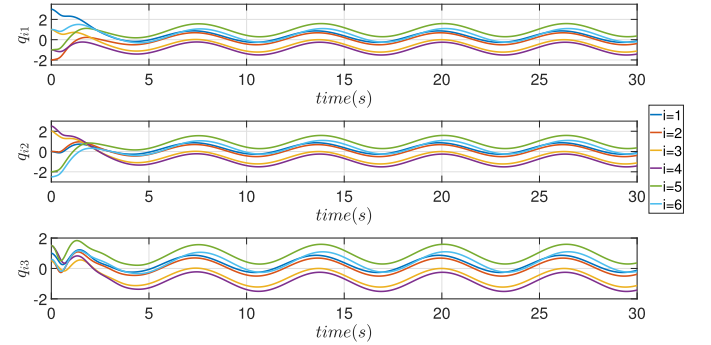


Fig. 14. The actual positions q_i in [42] under all link FDI attacks.

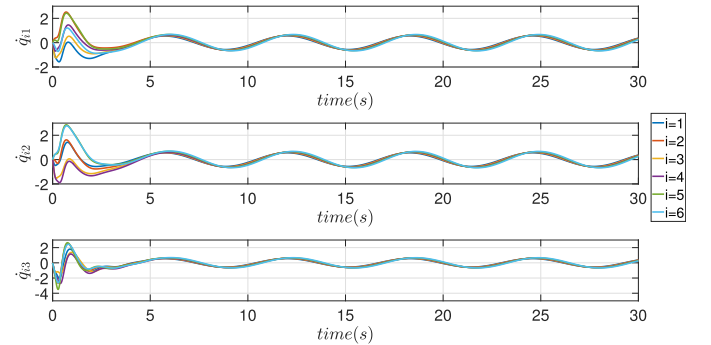


Fig. 15. The actual velocities $\dot{q}_i$ in [42] under all link FDI attacks

illustrated in Figs. 7 and 8. The position tracking errors e_i and velocities tracking errors $\dot{e}_i$ are illustrated in Figs. 9 and 10. Figs. 7–10 demonstrate that the quadrotor AAVs can maintain tracking within a narrow vicinity of the NE points, even when facing actuator failures due to physical attacks, highlighting the robustness of the proposed algorithm. The motion trajectories of the quadrotor AAVs in three-dimensional space are depicted in Fig. 11, where diamond symbols represent the initial states, and asterisks denote the final positions.

Furthermore, to demonstrate the disturbance rejection capability of our proposed algorithm, we also provide simulation results for the system's position and velocity using a controller without the disturbance rejection algorithm for comparison. From the comparison between Figs. 5–6 and Figs. 12–13, it is evident that in the system without the disturbance rejection term, all AAVs exhibit significant fluctuations in both position and

velocity along the x -axis and y -axis. In contrast, in the system with the designed disturbance rejection term, the positions and velocities of all AAVs converge to their desired values, with minimal fluctuation observed.

Remark 3: The conducted simulations primarily rely on synthetic data, designed to provide initial validation of the proposed algorithm. While this approach ensures consistency and control over experimental variables, and it inherently simplifies the complexity of real-world applications. To bridge this gap, we introduce model uncertainties and control input disturbances in the simulations, thereby mimicking practical flight conditions as closely as possible. The results demonstrate the algorithm's robustness to such disturbances, affirming its potential applicability in real-world scenarios.

Remark 4: The scalability of distributed algorithms in large-scale AAV clusters poses significant challenges due to the increased computational and communication demands. As the number of AAVs continues to grow, the computational and communication overhead will inevitably increase. To address these challenges, a hybrid event-triggered mechanism, combining controller-triggered and communication-triggered events, offers a promising solution for reducing unnecessary costs. This approach has been validated in similar scenarios [33], [34], [35], [36], [37] and can be incorporated into our future work to enhance its scalability.

To further validate the effectiveness of our algorithm in resisting composite attacks, we conducted a comparison with related studies, particularly [42], which addressed distributed AAV noncooperative games under specific network channel attacks and actuator failures caused by physical attacks. As illustrated in Figs. 14–15, the algorithm proposed in [42] achieves only bounded convergence under full-link communication FDI and computational FDI attacks, failing to enable the distributed AAV system to precisely reach the NE point. In contrast, as shown in Figs. 7–8, our algorithm exhibits superior robustness, particularly in mixed attack scenarios. It successfully achieves NE even under combined communication, computational, and physical attacks.

V. CONCLUSION

This paper investigates the resilience of distributed quadrotor AAVs in a noncooperative game, taking into account FDI attacks, physical attacks, uncertain dynamic parameters, and time-varying disturbances. In the game layer, a novel nonlinear attack observer is proposed to mitigate FDI attacks during communication and NE seeking. This enables all agents to converge to their respective NE points with high precision. In the execution layer, the impact of physical attacks and resulting actuator damage on quadrotor AAVs is addressed. A fault-tolerant algorithm is employed to rectify actuator failures, ensuring accurate tracking of the vicinity of NE points by all quadrotor AAVs. The asymptotic convergence of the proposed algorithm is demonstrated using Lyapunov stability analysis. Numerical simulations are conducted to validate the performance of the developed algorithm.

REFERENCES

- [1] G. Cui, H. Xu, X. Chen, and J. Yu, "Fixed-time distributed adaptive formation control for multiple QUAVs with full-state constraints," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 4, pp. 4192–4206, Aug. 2023.
- [2] H. Guo, M. Chen, Y. Jiang, and M. Lungu, "Distributed adaptive human-in-the-loop event-triggered formation control for QUAVs with quantized communication," *IEEE Trans. Ind. Inform.*, vol. 19, no. 6, pp. 7572–7582, Jun. 2023.
- [3] Y. Ouyang, L. Xue, L. Dong, and C. Sun, "Neural network-based finite-time distributed formation-containment control of two-layer quadrotor UAVs," *IEEE Trans. Syst., Man, Cybern., Syst.*, vol. 52, no. 8, pp. 4836–4848, Aug. 2022.
- [4] L. Quan et al., "Robust and efficient trajectory planning for formation flight in dense environments," *IEEE Trans. Robot.*, vol. 39, no. 6, pp. 4785–4804, Dec. 2023, doi: [10.1109/TRO.2023.3301295](https://doi.org/10.1109/TRO.2023.3301295).
- [5] M. Lv and N. Wang, "Distributed control for uncertain multiagent systems with the powers of positive-odd numbers: A low-complexity design approach," *IEEE Trans. Autom. Control*, vol. 69, no. 1, pp. 434–441, Jan. 2024.
- [6] D. Yu, S. Ma, Y. J. Liu, Z. Wang, and C. L. P. Chen, "Finite-time adaptive fuzzy backstepping control for quadrotor UAV with stochastic disturbance," *IEEE Trans. Automat. Sci. Eng.*, vol. 21, no. 2, pp. 1335–1345, Apr. 2024, doi: [10.1109/TASE.2023.3282661](https://doi.org/10.1109/TASE.2023.3282661).
- [7] M. Labbadi and M. Cherkaoui, "Adaptive fractional-order nonsingular fast terminal sliding mode based robust tracking control of quadrotor UAV with Gaussian random disturbances and uncertainties," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 57, no. 4, pp. 2265–2277, Aug. 2021.
- [8] H. Jiang, Q. Ma, and J. Guo, "Fuzzy-based fixed-time attitude control of QUAV with full-state constraints: Theory and experiments," *IEEE Trans. Fuzzy Syst.*, vol. 32, no. 3, pp. 1108–1115, Mar. 2024, doi: [10.1109/TFUZZ.2023.3318572](https://doi.org/10.1109/TFUZZ.2023.3318572).
- [9] C. Bai, P. Yan, H. Piao, W. Pan, and J. Guo, "Learning-based multi-UAV flocking control with limited visual field and instinctive repulsion," *IEEE Trans. Cybern.*, vol. 54, no. 1, pp. 462–475, Jan. 2024, doi: [10.1109/TCYB.2023.3246985](https://doi.org/10.1109/TCYB.2023.3246985).
- [10] W. Wang, L. Wang, J. Wu, X. Tao, and H. Wu, "Oracle-guided deep reinforcement learning for large-scale multi-UAVs flocking and navigation," *IEEE Trans. Veh. Technol.*, vol. 71, no. 10, pp. 10280–10292, Oct. 2022.
- [11] H. Qiu and H. Duan, "A multi-objective pigeon-inspired optimization approach to UAV distributed flocking among obstacles," *Inf. Sci.*, vol. 509, pp. 515–529, 2020.
- [12] N. Chen et al., "A self-rotating, single-actuated UAV with extended sensor field of view for autonomous navigation," *Sci. Robot.*, vol. 8, no. 76, 2023, Art. no. eade4538.
- [13] M. Ye, D. Li, Q. L. Han, and L. Ding, "Distributed Nash equilibrium seeking for general networked games with bounded disturbances," *IEEE/CAA J. Automatica Sinica*, vol. 10, no. 2, pp. 376–687, Feb. 2023.
- [14] Z. Feng, G. Hu, X. Dong, and J. Lü, "Adaptively distributed Nash equilibrium seeking of noncooperative games for uncertain heterogeneous linear multi-agent systems," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 6, pp. 3871–3882, Nov./Dec. 2023.
- [15] X. Shi, G. Wen, and X. Yu, "Finite-time convergent algorithms for time-varying distributed optimization," *IEEE Control Syst. Lett.*, vol. 7, pp. 3223–3228, 2023.
- [16] Y. Zhao, Q. Tao, C. Xian, Z. Li, and Z. Duan, "Prescribed-time distributed Nash equilibrium seeking for noncooperation games," *Automatica*, vol. 151, 2023, Art. no. 110933.
- [17] Y. Gao, J. Ma, J. Wang, and Y. Wu, "Event-triggered adaptive fixed-time secure control for nonlinear cyber-physical system with false data-injection attacks," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 70, no. 1, pp. 316–320, Jan. 2023.
- [18] J. Xiao and M. Feroskhan, "Cyber attack detection and isolation for a quadrotor UAV with modified sliding innovation sequences," *IEEE Trans. Veh. Technol.*, vol. 71, no. 7, pp. 7202–7214, Jul. 2022.
- [19] A. Upreti and D. B. Rawat, "Reinforcement learning for IoT security: A comprehensive survey," *IEEE Internet Things J.*, vol. 8, no. 11, pp. 8693–8706, Jun. 2021.
- [20] L. Dong, H. Xu, L. Zhang, Z. Li, and Y. Cheng, "Adjustable proportional-integral multivariable observer-based FDI attack dynamic reconstitution and secure control for cyber-physical systems," *Appl. Math. Comput.*, vol. 443, no. 15, 2023, Art. no. 127762.

- [21] W. Chung and H. Son, "Fault-tolerant control of multirotor UAVs by control variable elimination," *IEEE/ASME Trans. Mechatron*, vol. 25, no. 5, pp. 2513–2522, Oct. 2020.
- [22] X. Wang and S. Sun, "Incremental fault-tolerant control for a hybrid quad-plane UAV subjected to a complete rotor loss," *Aerosp. Sci. Technol.*, vol. 125, 2022, Art. no. 107105.
- [23] M. Saied, H. Shraim, and C. Francis, "A review on recent development of multirotor UAV fault-tolerant control systems," *IEEE Aerosp. Electron. Syst. Mag.*, vol. 39, no. 9, pp. 146–180, Sep. 2024, doi: [10.1109/MAES.2023.3327697](https://doi.org/10.1109/MAES.2023.3327697).
- [24] M. Lv, C. K. Ahn, B. Zhang, and A. Fu, "Fixed-time antisaturation cooperative control for networked fixed-wing unmanned aerial vehicles considering actuator failures," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 59, no. 6, pp. 8812–8825, Dec. 2023.
- [25] B. Wang, X. Peng, M. Jiang, and D. Liu, "Real-time fault detection for UAV based on model acceleration engine," *IEEE Trans. Instrum. Meas.*, vol. 69, no. 12, pp. 9505–9516, Dec. 2020.
- [26] X. Gong, M. V. Basin, Z. Feng, T. Huang, and Y. Cui, "Resilient time-varying formation-tracking of multi-UAV systems against composite attacks: A two-layered framework," *IEEE-CAA J. Automatica Sinica*, vol. 10, no. 4, pp. 969–984, Apr. 2023.
- [27] K. Liu, R. Wang, X. Wang, and X. Wang, "Anti-saturation adaptive finite-time neural network based fault-tolerant tracking control for a quadrotor UAV with external disturbances," *Aerosp. Sci. Technol.*, vol. 115, 2021, Art. no. 106790.
- [28] D. T. Nguyen, D. Saussie, and L. Saydy, "Design and experimental validation of robust self-scheduled fault-tolerant control laws for a multicopter UAV," *IEEE/ASME Trans. Mechatron*, vol. 26, no. 5, pp. 2548–2557, Oct. 2021.
- [29] M. Ye and G. Hu, "Distributed Nash equilibrium seeking by a consensus based approach," *IEEE Trans. Autom. Control*, vol. 62, no. 9, pp. 4811–4818, Sep. 2017.
- [30] A. R. Romano and L. Pavel, "Dynamic NE seeking for multi-integrator networked agents with disturbance rejection," *IEEE Trans. Control Netw. Syst.*, vol. 7, no. 1, pp. 129–139, Mar. 2020.
- [31] H. Duan, L. Pei, and Y. Yu, "A predator-prey particle swarm optimization approach to multiple UCAV air combat modeled by dynamic game theory," *IEEE/CAA J. Automatica Sinica*, vol. 2, no. 1, pp. 11–18, Jan. 2015.
- [32] Z. Zhang, J. Jiang, H. Xu, and W. A. Zhang, "Distributed dynamic task allocation for unmanned aerial vehicle swarm systems: A networked evolutionary game-theoretic approach," *Chin. J. Aeronaut.*, vol. 37, no. 6, pp. 182–204, 2024.
- [33] D. Wang, Z. Wang, Z. Wang, and W. Wang, "Design of hybrid event-triggered containment controllers for homogeneous and heterogeneous multiagent systems," *IEEE Trans. Cybern.*, vol. 51, no. 10, pp. 4885–4896, Oct. 2021.
- [34] H. Liang, Z. Chang, and C. K. Ahn, "Hybrid event-triggered intermittent control for nonlinear multi-agent systems," *IEEE Trans. Netw. Sci. Eng.*, vol. 10, no. 4, pp. 1975–1984, Jul./Aug. 2023.
- [35] G. Zhao, H. Cui, and C. Hua, "Hybrid event-triggered bipartite consensus control of multiagent systems and application to satellite formation," *IEEE Trans. Automat. Sci. Eng.*, vol. 20, no. 3, pp. 1760–1771, Jul. 2023.
- [36] G. Zhao, C. Hua, and X. Guan, "A hybrid event-triggered approach to consensus of multiagent systems with disturbances," *IEEE Trans. Control Netw. Syst.*, vol. 7, no. 3, pp. 1259–1271, Jul. 2020.
- [37] M. Hua, H. Ding, X. Yao, and W. Liu, "Distributed fixed-time Nash equilibrium seeking algorithm for multiple ASVs: A hybrid event-triggered approach," *Ocean Eng.*, vol. 290, 2023, Art. no. 116410.
- [38] P. Frihauf, M. Krstic, and T. Basar, "Nash equilibrium seeking in noncooperative games," *IEEE Trans. Autom. Control*, vol. 57, no. 5, pp. 1192–1207, May 2012.
- [39] L. J. Ratliff, S. A. Burden, and S. S. Sastry, "On the characterization of local Nash equilibria in continuous games," *IEEE Trans. Autom. Control*, vol. 61, no. 8, pp. 2301–2307, Aug. 2016.
- [40] Z. Deng, J. Luo, Y. Liu, and W. Yu, "Distributed formation control algorithms for QUAVs based on aggregative games," *IEEE Syst. J.*, vol. 17, no. 3, pp. 4419–4429, Sep. 2023.
- [41] W. Liu, X. Yao, M. Ge, M. Hua, and H. Ding, "Hierarchical Nash equilibrium seeking strategies of quadratic time-varying games with Euler–Lagrange players," *Int. J. Robust Nonlinear Control*, vol. 34, no. 1, pp. 666–680, 2024.
- [42] M. J. Geng, H. F. Ding, X. Y. Yao, W. J. Liu, and M. Hua, "Noncooperative game of distributed quadrotor UAVs with multiple constraints," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 60, no. 4, pp. 4728–4739, Aug. 2024.



Mei-Jie Geng received the B.S. degree in mechanical engineering in 2021 from the China University of Geosciences, Wuhan, China, where he is currently working toward the Ph.D. degree in mechanical engineering with the School of Mechanical Engineering and Electronic Information. His research interests include distributed control, game theory, and reinforcement learning control, with applications in multi-agent systems and autonomous autonomous systems.



Hua-Feng Ding received the B.S., M.S., and Ph.D. degrees in mechanical engineering from Yanshan University, Qinghuadao, China, in 2000, 2003, and 2007, respectively, and the second Ph.D. degree in mechanical engineering from the University of Duisburg-Essen, Essen, Germany, in 2015. From 2010 to 2012, he was an Alexander von Humboldt Research Fellow with the University of Duisburg-Essen. He is currently a Professor with the China University of Geosciences, where he is also the President with the School of Mechanical Engineering and Electronic Information. His research interests include robotics, mechatronics, innovative design, and control of mechanical equipments.



Xiang-Yu Yao received the B.Eng. and Ph.D. degree in mechanical engineering from the China University of Geosciences, Wuhan, China, in 2015 and 2021, respectively. He was a Visiting Scholar with the Department of Electrical Engineering, Yeungnam University, Gyeongsan, South Korea, from 2019 to 2020, and a Postdoctoral Fellow with the School of Automation, China University of Geosciences, Wuhan, China, from 2021 to 2024. He is currently a Professor with the China University of Geosciences. His research interests include optimization control, noncooperative games, and learning-based autonomous systems.



Maolong Lv received the Ph.D. degree from the Delft Center for Systems and Control, Delft University of Technology, Delft, The Netherlands, in 2021. He is a national-level Young Talent in China and is currently with Air Force Engineering University, Xi'an, China. His research interests include adaptive learning control, distributed control, reinforcement learning, and intelligent decision-making with applications in AAVs, intelligent air combat, and man-autonomous teaming. Dr. Lv was awarded a Descartes Excellence Fellowship from the French Government in 2018, which allowed him a research visit from 2018 to 2019 with the University of Grenoble.